



Personal Budget Tracking System

Name : Sameer S
Reg no. : 727723EUCB49

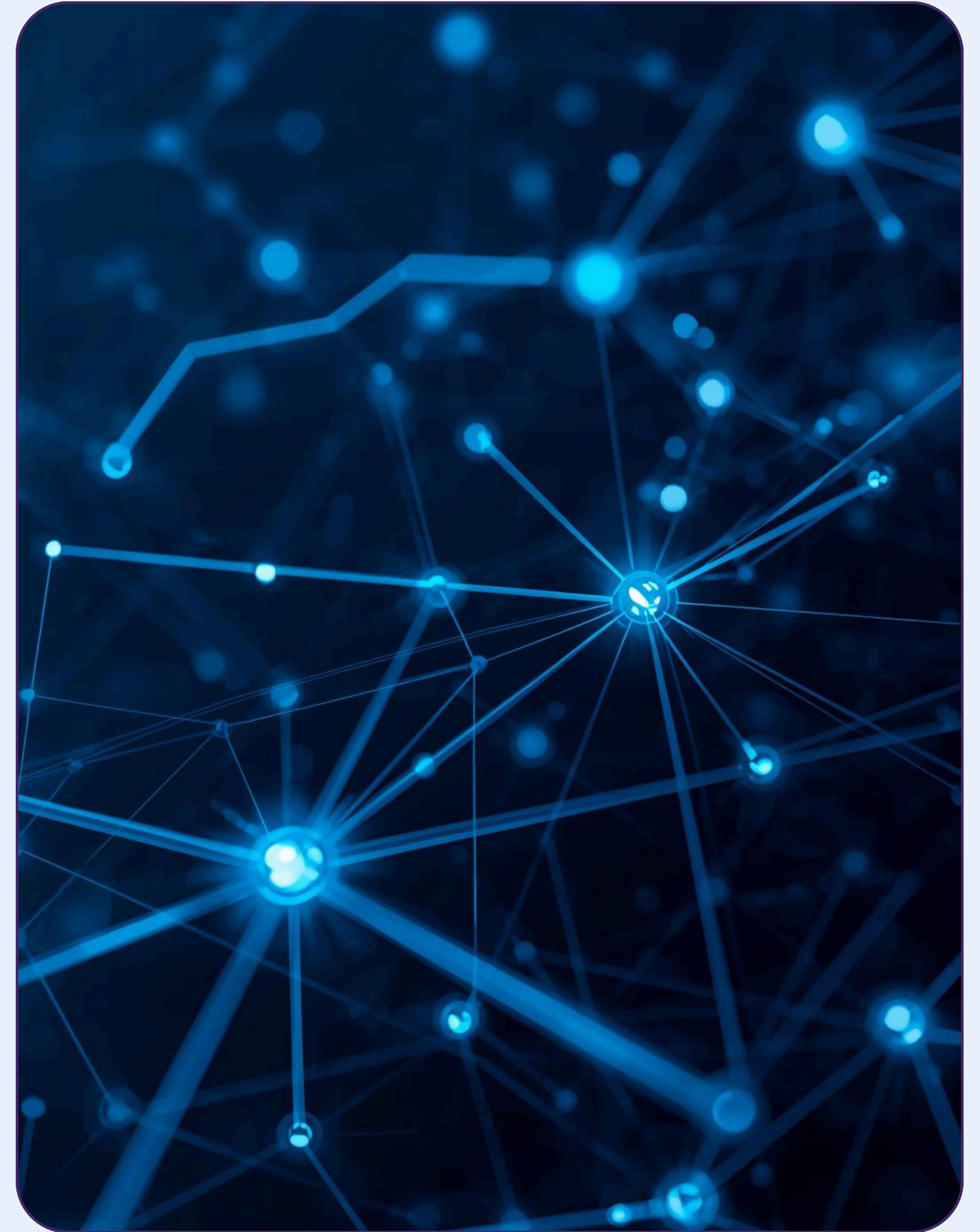


Overview

Many individuals struggle to track where their money goes each month, often resulting in overspending or weak savings habits.

The Personal Budget Tracking System is designed to make this process easier by providing a simple digital platform to record income, categorize expenses, and monitor spending patterns in real time. It eliminates the need for manual calculations and helps users understand their financial habits better.

By offering clear insights and visual summaries, the system allows users to identify unnecessary expenses, plan budgets more effectively, and make informed financial decisions.





Problem Statement

Many people struggle to maintain control over their finances because they lack a clear and consistent way to track income and expenses.

As a result, they often:

- Fail to track expenses properly, leading to confusion about where their money goes.
- Overspend without realizing it, which affects their ability to save effectively.
- Rely on manual calculations, which are time-consuming and prone to human error.

The Personal Budget Tracking System aims to overcome these challenges by introducing automation and real-time expense tracking, ensuring users can easily manage and understand their financial activities.

Objectives

Expense Monitoring: Record and categorize expenses efficiently.

Budget Planning: Help users allocate budgets and track deviations.

Financial Insights: Visualize spending patterns through reports and charts.

Data Security: Ensure all personal and financial information remains private and protected.

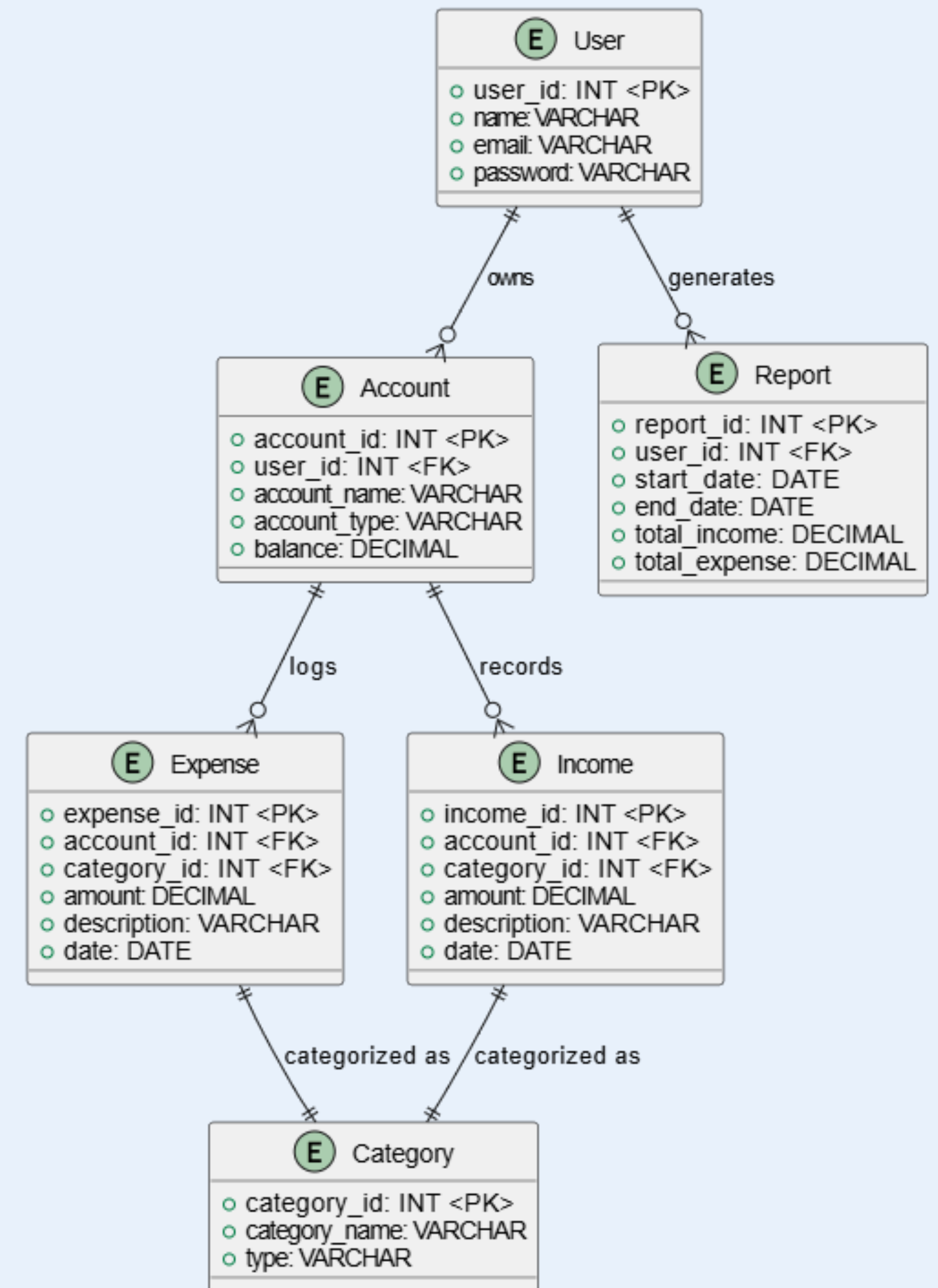


ER Diagram

The Entity-Relationship (ER) Diagram represents the logical structure of the system's database.

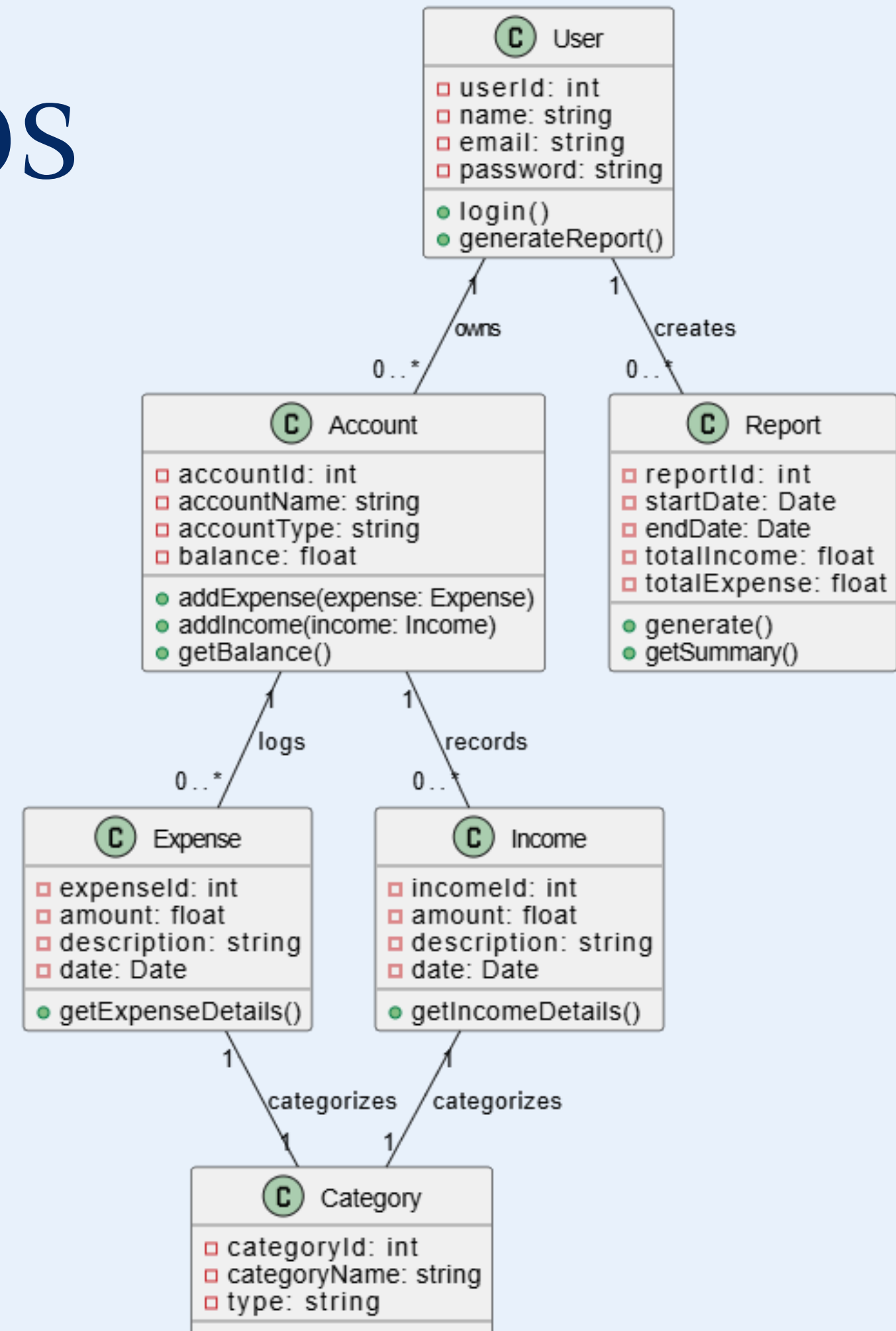
It shows how different entities, such as Users, Expenses, Income, and Categories, are connected and interact with each other.

This diagram helps visualize how data flows within the system, for example, a user can record multiple income and expense entries, each linked to specific categories.



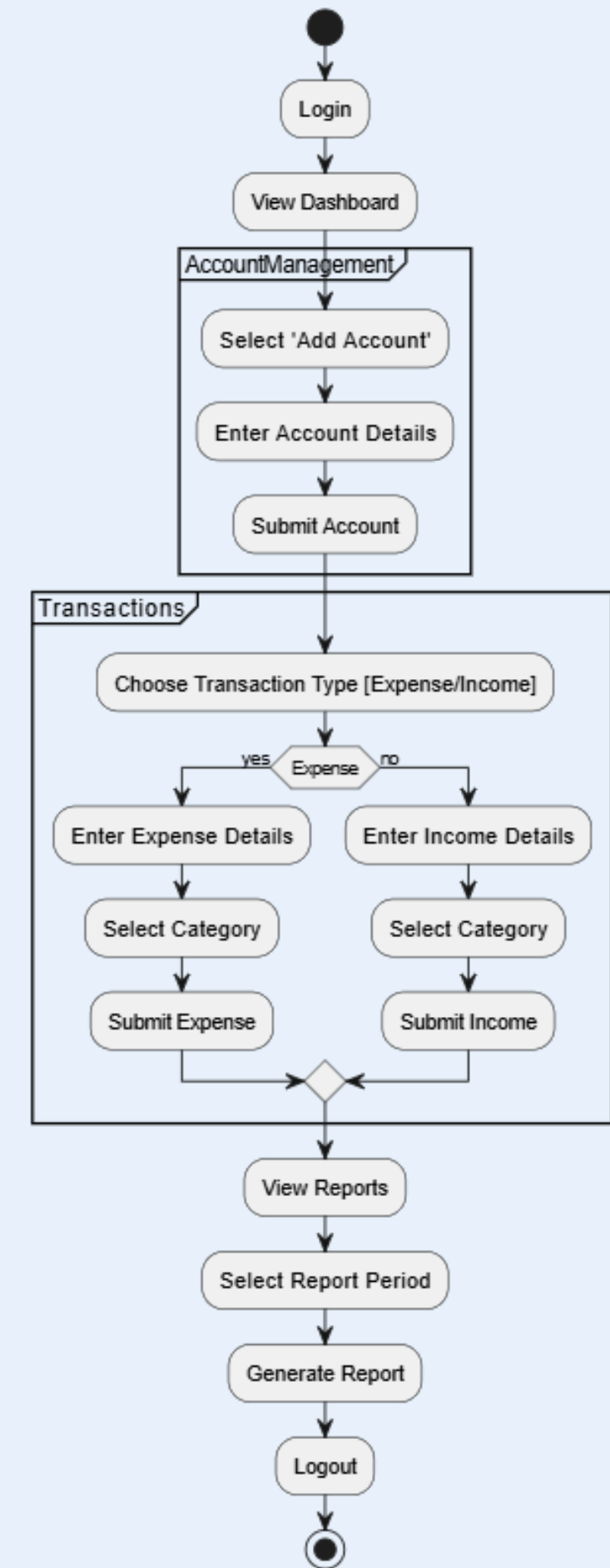
Class Relationships

- Shows the logical structure of the database and data flow.
- Entities: User, Account, Income, Expense, Category
- Relationships:
 - User → multiple Accounts, Income, and Expenses
 - Income/Expense → linked to a Category
 - Accounts → track all transactions
- Purpose: Visualizes how data is connected for accurate tracking and reporting



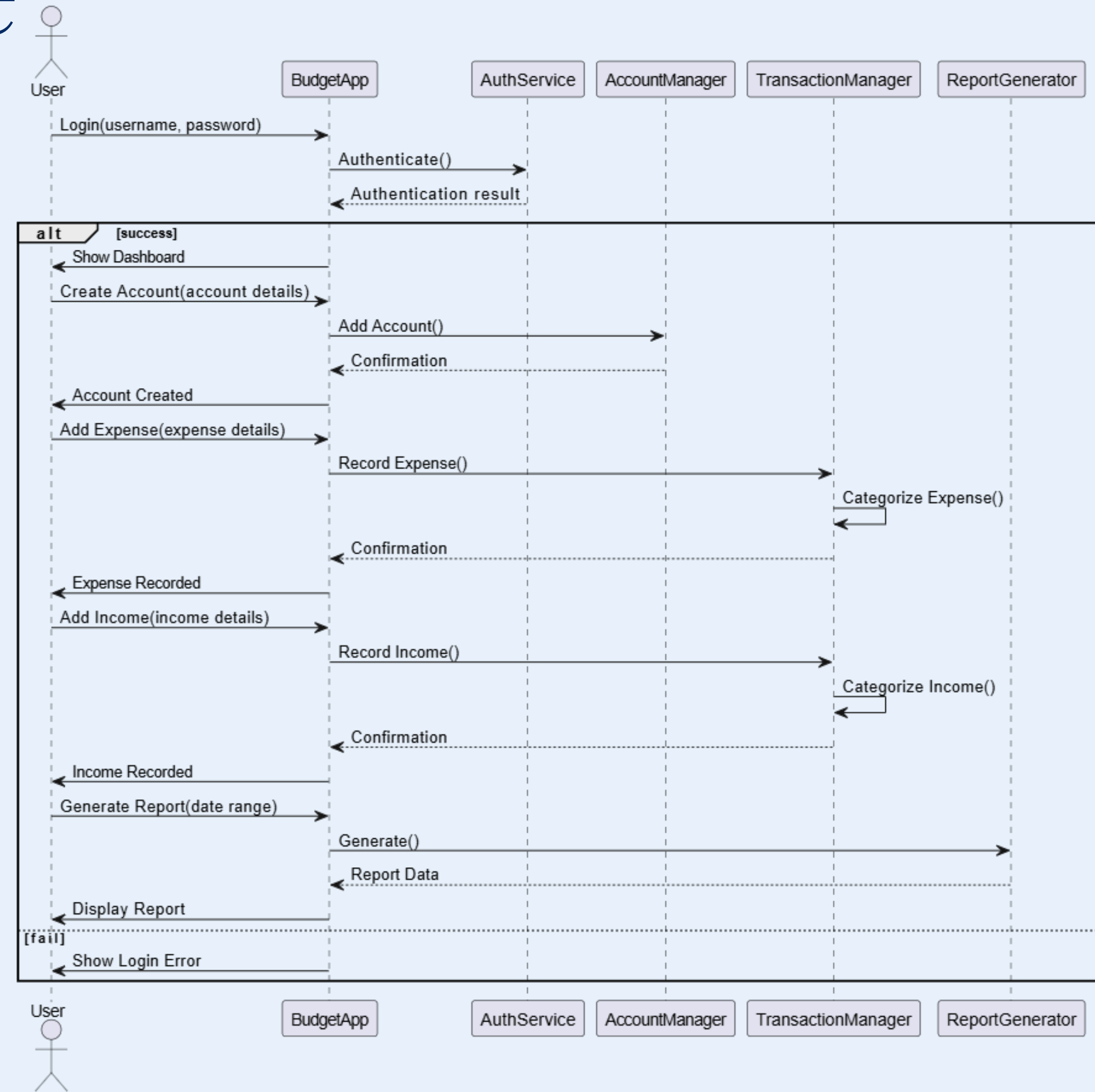
Activity Diagram

- Shows the flow of user actions in the system.
- Main Activities:
 - a. Login/Register
 - b. Manage Accounts
 - c. Record Transactions (Income/Expense)
 - d. Set Budgets
 - e. Generate Reports
 - f. Logout
- Purpose: Visualizes step-by-step user interactions and system workflow.



System Sequence

- **Purpose:** Illustrates the interaction between the user and the system over time for key processes.
- **Key Scenarios:**
 1. Add Expense:
 - User enters expense → System verifies category → Updates account balance → Confirms to user
 2. Generate Report:
 - User requests report → System fetches transactions/budget → Displays summary and insights
- **Benefits:**
 - Visualizes step-by-step flow of operations
 - Helps understand system behavior and message sequence



Backend Logic

The backend is responsible for managing all core functionalities:

CRUD Operations: Create, Read, Update, Delete financial data.

User Authentication: JWT-based secure login system.

Data Validation: Ensures accurate inputs.

Report Generation: Aggregates data for insights.
A well-structured backend improves reliability, speed, and scalability.





Technology Stack

Frontend: React / HTML / CSS / JavaScript – for a responsive and interactive user interface.

Backend: Flask / Node.js – for handling server-side logic and data processing.

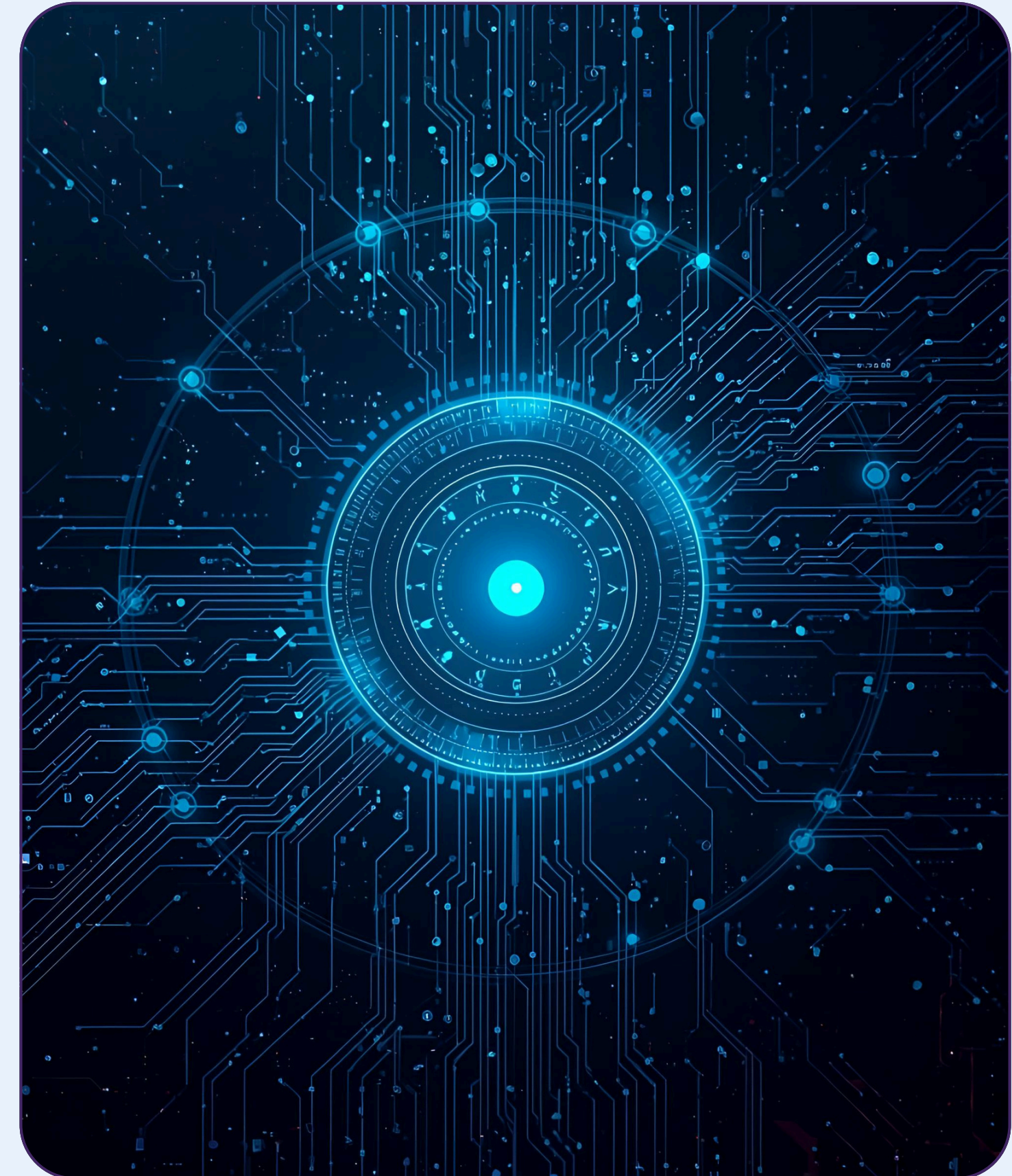
Database: MySQL / MongoDB – to store transaction data securely.

Authentication: JWT / OAuth 2.0 – for secure login and user management.

Hosting: AWS / Firebase – for scalable and reliable deployment.

System Architecture

- The system architecture outlines the interaction between the user interface, backend server, and database.
- The user interacts with the web interface to add or view transactions.
- Requests are sent to the backend via REST APIs.
- The backend validates, processes, and stores data in the database.
- Reports and insights are generated dynamically for the user.



Conclusion

The Personal Budget Tracking System provides users with a secure and efficient way to manage their finances.

It promotes better financial habits through data-driven insights and automation.

The project demonstrates the importance of integrating user-friendly design with a robust backend to achieve real-world financial control.